

Operator Discovery Report: scaffold_selector_transfer_focu S

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Select a compact baseline scaffold per instance structure to improve held-out transfer, emphasizing robustness on bottleneck/two-cluster shapes and clustered Euclidean instances; otherwise fall back to nearest_neighbor_2
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:random_restart_2opt.
- Surviving candidate: False.
- Problem class where it helps: clustered_tsp, grid_like_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: d2256b0bb0be1564.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on heldout_tsplib with gap delta 0.01955.
- Underperforms on elongated_corridor_tsp with gap delta 0.034174.
- Underperforms on nearest_neighbor_trap_tsp with gap delta 0.3987.
- Underperforms on two_cluster_bottleneck_tsp with gap delta 0.022965.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into nearest_neighbor_2opt.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Does not transplant cleanly into random_restart_2opt.
- Does not transplant cleanly into sparse_three_opt.
- Disabling the operator does not hurt, so the ablation does not support a real operator effect.
- Runtime inflation is high at 1.679113.

Transplant Results

- nearest_neighbor_2opt gap delta 0.02022, runtime delta 321.779129, distance-eval delta 11882.714286.
- cheapest_insertion_2opt gap delta 0.127541, runtime delta -4654.465843, distance-eval delta -6749175.285714.
- random_restart_2opt gap delta 0.040545, runtime delta 140.9561, distance-eval delta -5278.285714.
- sparse_three_opt gap delta 0.021505, runtime delta 382.081543, distance-eval delta 8461.428571.
- clustered_local_search gap delta 0.00245, runtime delta 215.5765, distance-eval delta -2549.0.

Instance-Family Results

- heldout_tsplib operator gap 0.254608 vs baseline 0.235058.
- clustered_tsp operator gap 0.053865 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.037378 vs baseline 0.003204.
- grid_like_tsp operator gap 0.082744 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.3987 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.132022 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: 0.042667.
- Runtime inflation: 1.679113.
- Distance-evaluation inflation: 2.049916.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator -0.043048.
- flattened_selector mean gap 0.121317 and delta vs full operator -0.043048.
- shuffled_selector mean gap 0.100777 and delta vs full operator -0.063588.